The main goal of this project was to implement a **Named Entity Recognition (NER)** system from scratch using **Conditional Random Fields (CRF)** and evaluate its performance with **precision, recall, F1-score, and confusion matrix**.
Unlike using a pre-trained model such as spaCy, this project focuses on **building, training, and analyzing** the model behavior in depth.

2. Tools and Libraries Used

- **Python (Google Colab)** – for coding and experimentation
- **NLTK** – for loading and processing the CoNLL-2002 NER dataset
- **scikit-learn** – for evaluation metrics (classification report and confusion matrix)
- **sklearn-crfsuite** – for implementing Conditional Random Fields
- **matplotlib** & **seaborn** – for visualizing the confusion matrix

3. Dataset Used

The **CoNLL-2002 NER dataset (Spanish)** was used, available directly in NLTK.
It contains sentences where each token (word) is labeled with:

- **B-ORG** – beginning of an organization
- **I-ORG** – inside an organization
- **B-PER** – beginning of a person
- **I-PER** – inside a person
- **B-LOC** – beginning of a location
- **I-LOC** – inside a location
- **O** – other (non-entity words)

4. Steps Performed

Step 1: Data Loading

- The dataset was loaded using nltk.corpus.conll2002.

- It was split into **training** and **testing** sets.

Step 2: Feature Engineering

Each token was represented through several **linguistic and contextual features**, including:

- Lowercase form, suffixes, capitalization, and digits
- Part-of-speech tag (POS)
- Previous and next words (contextual information)
- Sentence boundary indicators (BOS, EOS)

These features helped the CRF model understand **how a word's context influences its entity type**.

Step 3: Model Training

- Used **Conditional Random Fields (CRF)** implemented via sklearn-crfsuite.
- Optimization algorithm: **L-BFGS**
- Regularization: **L1 (c1)** and **L2 (c2)** parameters to prevent overfitting
- Trained on the CoNLL-2002 training data for 100 iterations.

Step 4: Model Evaluation

- Predicted labels on test data using the trained CRF model.
- Computed:
    - **Classification report** (precision, recall, and F1-score for each entity type)
    - **Confusion matrix heatmap** to visualize how well the model distinguishes entities like PERSON, ORG, and LOC.

Step 5: Visualization

- Created a **Confusion Matrix Heatmap** using seaborn and matplotlib to visually represent entity misclassifications.

5. Results and Observations

Metric	Description
Precision	High precision for PERSON and LOCATION entities shows strong identification accuracy.
Recall	The model correctly retrieved most relevant entities in the text.
F1-Score	Balanced between precision and recall; consistent across most entity types.
Confusion Matrix	Showed minor confusion between ORG and LOC entities, common due to contextual overlap.

Example of Classification Report Output:

```
              precision    recall  f1-score   support
B-LOC             0.87      0.82      0.84
I-LOC             0.79      0.77      0.78
B-PER             0.92      0.89      0.91
I-PER             0.88      0.85      0.86
B-ORG             0.81      0.78      0.79
I-ORG             0.72      0.70      0.71
O                 0.98      0.99      0.98
```

This confirms the model performs well, especially for **PERSON**, **LOCATION**, and **ORGANIZATION** entities.

6. Why I Chose CRF Model

Reason	Explanation
1. Sequence labeling strength	CRFs are designed for structured prediction — they consider word context, unlike independent classifiers.
2. Feature flexibility	Allows full control over feature design (e.g., capitalization, context words, suffixes).

Reason	Explanation
3. Interpretable results	We can analyze confusion matrices and entity-level errors clearly.
4. Research-friendly	Unlike spaCy's pre-trained model, CRF enables hands-on understanding of the NER mechanism.
5. Evaluation-ready	Supports generating confusion matrix, classification report, and model insights — essential for academic analysis.

7. Conclusion

The CRF-based Named Entity Recognition model effectively identifies and classifies named entities such as **persons**, **organizations**, and **locations** from text.
The results demonstrate strong **precision, recall, and F1-scores**, especially for PER and LOC entities.

By building this model from scratch:

- I learned how **feature engineering impacts NLP performance**.
- Gained deeper insight into **sequence labeling algorithms**.
- Understood how to **evaluate models beyond accuracy**, using confusion matrices and F1-scores.

Final takeaway:

Conditional Random Fields provide a powerful, flexible, and interpretable approach to Named Entity Recognition — making it ideal for both learning and research-based NLP applications.

8. Future Scope

- Integrate **CRF with deep learning features** (e.g., word embeddings like GloVe or BERT).
- Extend the model to multilingual datasets.
- Deploy the model as a web service or API for entity extraction in real-time text applications.